

Architecture Decision Record

Hybrid Multilingual Retrieval, Reranking, and Resilient Grounded Generation

Status: Accepted

Date: 2026-10-02

Project: Yemen Opportunity Navigator

Context

Yemen Opportunity Navigator answers Arabic and English questions about opportunities using a curated source corpus while keeping answers grounded in source evidence. The validated implementation contains 49 processed source records and 164 searchable chunks. The architecture must support semantic matching, exact-term matching, source attribution, measurable retrieval quality, reproducible evaluation, and graceful behavior when external AI services are unavailable.

Decision

1. Cohere embed-multilingual-v3.0 for document and query embeddings.
2. Persistent Chroma for semantic vector retrieval.
3. BM25 with Arabic/English normalization for lexical retrieval.
4. Hybrid Vector + BM25 fusion using RRF to form a broader candidate set.
5. 20 hybrid candidates passed to Cohere rerank-multilingual-v3.0.
6. Top 5 reranked chunks used as the grounded generation context.
7. Command A (command-a-03-2025) generates one grounded draft by default; the second draft-plus-context review is optional via `RAG_ENABLE_ANSWER_REVIEW=1`.
8. Source attribution is built separately from answer text, then rendered with the final answer.
9. Streamlit UI with Supabase authentication/session management; identical public queries are cached for 5 minutes to reduce repeated API calls.

Production Reliability Update

The public deployment defaults to one Chat call per uncached query. Monthly/trial quota exhaustion is treated as non-retryable; only transient provider/network failures are retried. Vector-search failure falls back to BM25, reranker failure keeps the existing hybrid/BM25 ordering, and generation failure returns retrieved official sources with a user notice instead of failing the search.

Alternatives Considered

Vector-only retrieval was simpler and strong at 96.67% Recall@5, but missed one golden question.
BM25-only retrieval preserved exact lexical matching but reached only 80.00%.
Hybrid without reranking broadened candidate diversity but achieved 90.00%, below vector-only performance at the final top-five level.
The accepted design therefore keeps hybrid retrieval for candidate breadth and learned reranking for final precision and recall.

Supporting Evidence

docs/technical_evidence.md
docs/charts/architecture_diagram.png
evaluation/recall_summary.md
evaluation/ragas_summary.md
evaluation/investor_cost_summary.md
cost_analysis.md

Evidence - Retrieval

Configuration	Recall@5
Vector only	96.67% (29/30)
BM25 only	80.00% (24/30)
Hybrid + RRF	90.00% (27/30)
Hybrid + Cohere Reranker	100.00% (30/30)

The decisive improvement comes from reranking the broader hybrid candidate set, not from raw hybrid fusion alone. Final Recall@5 is also 100% for Arabic (15/15), English (15/15), and the hard subset (11/11).

Evidence - Generation Quality

RAGAS metric	Score
Faithfulness	0.9875
Answer Relevancy	0.8227
Context Precision	1.0000
Context Recall	1.0000

The project-level arithmetic mean across these four metric averages is 0.9526; it is a project summary, not a separate canonical RAGAS metric.

Operational Trade-off

The reported RAGAS and cost-profile measurements were produced with the evaluated two-pass configuration: 2 Chat calls + 1 Embed call + 1 Rerank call per normal query. Across 30/30 successful cost-profile queries:

- Mean Command A generation cost: \$0.01982967/query
- P95 generation cost: \$0.02205750/query
- P50 end-to-end latency: 10.945 s
- P95 latency: 32.381 s

Current public default: 1 Chat + 1 Embed + 1 Rerank for an uncached normal query; the review pass can be re-enabled to reproduce the evaluated two-pass path.

Consequences

Benefits: 100% Recall@5 on the current golden set, bilingual retrieval robustness, excellent context precision/recall, high faithfulness, explicit source attribution, and graceful degradation.
Costs: reranking adds API usage and latency; enabling the optional review pass adds a second generation call.
Known weakness: Answer Relevancy at 0.8227 is the primary measured quality improvement area.
Limitations: results are based on 30 retrieval questions, 20 RAGAS questions, and the current 49-source/164-chunk corpus; they are not guarantees for all future queries or larger deployments.
OCR/VLM: not in the default ingestion path because current sources are machine-readable; conditional OCR/VLM is planned for scanned pages, complex tables, figures, graphs, and other visual content.